

spot chekers lis alt proof

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Lemma 0.1. *an element i being 0 big can be detected with probability $\frac{3}{4}$ in $O(\log n)$ time.*

Proof. wlog let j be the smallest witness to i 's bigness and $j > i$. There exists some n such that $j \in [i + 2^n, i + 2^{n+1}]$. Since j is the smallest witness, the number of violations that $i + 2^n$ sees is $< \frac{1}{2}$. the number of violations that $i + 2^{n+1}$ witnesses is at least $\frac{j-i+1}{2} \geq \frac{1}{4} \cdot 2^{n+1}$. The probability of finding a violation given n samples is $(1 - \frac{3}{4})^n$. If n is large enough, the probability that arbitrarily picking an element k in $[i + 2^n, i + 2^{n+1}]$ a violation $f(k) < f(i)$ is found is greater than $\frac{3}{4}$. Doing this procedure on i for all n in both directions until indecies exceed the bounds of the array guarantees that this algorithm returns BIG in $2 \log_2 n$ time with at least $\frac{3}{4}$ probability. $\square$

Lemma 0.2. *Given an array of booleans of length n with ε Trues, the probability of picking at least 1 True given m samples of the uniform distribution is $\geq 1 - (e^{-m\varepsilon/2})$.¹*

Proof. We know

$$P(X \leq (1 - \delta)\mu) \leq e^{-\frac{\mu\delta^2}{2}} \quad (1)$$

by the Chernoff bound formula. The expected number of times True is picked given m samples is $m\varepsilon$. By setting $\delta = 1$ and X to be the sum of m outcomes of Bernoulli distribution with success ε , we see the probability of none of the Bernoulli distributions returning 1 (i.e. sampling true) is

$$P(X = (1 - 1)\mu = 0) \leq e^{-\frac{m\varepsilon^2}{2}} \quad (2)$$

The probability of picking at least 1 True is 1 minus this value.²

$\square$

Lemma 0.3. *Given $a < b \in [n]$ and $f(a) > f(b)$, either a or b is 0 big.*

Proof. let $c \in (a, b)$. $f(c)$ is either

1. greater than $f(a)$

¹this is independent of n (!!)

²The RHS of the can be set arbitrarily small by setting $m = O(\frac{1}{\varepsilon})$.

2. less than $f(a)$ and greater than $f(b)$
3. less than $f(b)$

We count the number of violations a witnesses for b , and b witnesses for a . The proof would follow if either a is the bigness witness for b or b is the bigness witness for a . Respective to the enumeration, c is in violation of b , a and b , a in each case respectively. Since every c is assigned as a violation for either a or b , by pigeonhole there must be at least $\frac{b-a+1}{2}$ violations for either a or b . $\square$

1 The algorithm

Given $f : [n] \rightarrow \mathbb{R}$

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for  $l$  do  $O(\frac{1}{\varepsilon})$  times:
    pick element  $a \in [n]$  and check if  $a$  is 0 big using lemma 0.1
    if 0 big then
        return FAIL
return PASS

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The probability detecting an element a is 0 big given it is in fact 0 big should be set to be $\frac{3}{4}i$ and the number of times the for loop repeats should be set so that the probability of sampling a 0 big element is at least i .

If εn elements must be removed for the function f to be monotone, then there must exist at least εn 0 big elements. This is true because an element e in violation of monotonicity implies e either works as a or b in lemma 0.3.

$$\begin{aligned}
 &= P(\text{FAIL} | \varepsilon \text{ violations of monotonicity}) \\
 &= P(\text{FAIL} | \varepsilon \text{ 0 big elements}) \\
 &= P(\text{FAIL} | \text{test 0 big element}) \cdot P(\text{test 0 big element} | \varepsilon \text{ 0 big elements})
 \end{aligned} \tag{3}$$

The final line is equal to greater than $\frac{3}{4}$.

NOTES: ignore

$m\varepsilon$ is the expected value

0 big can be checked in $\log n$ time. candidate 2^n being witness for i being $1/4$ big is checked in $O(1)$ (this overflows but that's ok) assume that i is 0-big. check s elements in interval $[i, i + 2^n]$. the probability that a witness j in the interval $[i + 2^{n-1}, i + 2^n]$ is NOT found is $(3/4)^s$. making s large makes the probability that the algo fails to find 0-big very small. note s is $O(1)$.

given the sequence should fail, the expected number of violations sampled is 1.

$P(\text{output F} \mid \text{bad sequence}) = 3/4$

$P(\text{output F} \mid \text{test bad element}) = 1 - \delta P(\text{test bad element} \mid \text{bad sequence})$
 $= \varepsilon$

$P(A) = P(A \mid B) P(B)$

assume: ε violate monotonicity

check $1/\varepsilon$ times. check for 0 big. if not found, continue. else fail.

pass

Lemma 1.1. *if an element i is in a subset s such that when $n - s$ gives an optimal monotone subsequence, then i is 0 big.*

Proof. i must have at least one witness j such that $i < j, f(i) > f(j)$ or $i > j, f(i) < f(j)$. $\square$

Lemma 1.2. *given an element in a list length n and that is $\varepsilon_f n$ away from monotonicity there exists an algorithm detects its violation from monotonicity that runs in $\log n$ time*

Proof. $\square$